

Credit Portfolio Calculation

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1 Introduction

This document presents various calculations methods for the loss distribution of a given credit portfolio under a Gaussian copula credit portfolio model. With some change, the methods presented here are also applicable if other copula functions are used.

2 Gaussian Copula Model

Consider a credit portfolio of n names, each of them has a survival time τ_i whose distribution is $F_i(t)$, and density function is $f_i(t)$. The joint survival time distribution is given as follows:

$$F(t_1, t_2, \dots, t_n) = \Phi_n(\Phi^{-1}(F_1(t_1)), \Phi^{-1}(F_2(t_2)), \dots, \Phi^{-1}(F_n(t_n)), \Sigma), \quad (1)$$

where Σ is the correlation matrix of the standardized asset returns $X_1, X_2, \dots, X_n$ of the n credits. There is a one-to-one mapping between the survival time τ_i and the asset return X_i , given by

$$\tau_i = F_i^{-1}(\Phi(X_i)) \quad (2)$$

3 Portfolio Loss Distribution

The total loss $L(t)$ up to time t for a portfolio of n credits is:

$$L(t) = \sum_{i=1}^n N_i(1 - R_i)I_{\tau_i \leq t} \quad (3)$$

where:

- N_i : Notional amount of the i -th credit.
- R_i : Recovery rate of the i -th credit.
- τ_i : Survival time for the i -th credit.
- $I_{\tau_i \leq t}$: Indicator function (1 if $\tau_i \leq t$, 0 otherwise).

4 Monte Carlo Simulation

Monte Carlo simulation can be implemented in Python, R, or C++. The key steps include:

- Generate correlated normal variables using Cholesky decomposition on the correlation matrix Σ .
- Compute default time based on the formula (2).

- Use the formula (3) for the total loss distribution.

This method is good with any correlation matrix, the drawbacks are very slow, less accurate with a small number of simulation runs.

5 Analytic Method with one Correlation Parameter

Analytic method is based on a simple assumption for the correlation matrix. We often make the following assumption: pairwise constant correlation, i.e. $\rho_{ij} = \rho$ for all i and j . In this case, we have a simple factor representation for the standardized asset return as follows: For each entity i in the portfolio ($i = 1, \dots, n$):

$$X_i = \sqrt{\rho} \cdot X_M + \sqrt{1 - \rho} \cdot \epsilon_i \quad (4)$$

where:

- X_i : Latent variable for credit i 's standardized asset return
- X_M : Systemic market risk factor $\sim \mathcal{N}(0, 1)$
- ϵ_i : Idiosyncratic risk factor $\sim \mathcal{N}(0, 1)$
- ρ : Correlation parameter ($0 \leq \rho \leq 1$)

This is usually called *one-factor* model¹ as the correlation comes only from a common factor X_M . Once we condition on this common factor, the asset returns or the survival times for all credits in the portfolio becomes *independent*. Then, we can use some analytic approaches for the portfolio loss calculation.

5.1 Conditional Default Probability

First, we calculate the conditional default probability on the common factor X_M for each credit. For simplicity, we also omit i in the following derivation.

$$\begin{aligned} q(t|X_M) &= \Pr[\tau \leq t|X_M] \\ &= \Pr[F^{-1}(\Phi(X)) \leq t|X_M] \\ &= \Pr[X \leq \Phi^{-1}(F(t))|X_M] \\ &= \Pr[\sqrt{\rho} \cdot X_M + \sqrt{1 - \rho} \cdot \epsilon \leq \Phi^{-1}(F(t))|X_M] \\ &= \Pr\left[\epsilon \leq \frac{\Phi^{-1}(F(t)) - \sqrt{\rho} \cdot X_M}{\sqrt{1 - \rho}}|X_M\right] \\ &= \Phi\left(\frac{\Phi^{-1}(q(t)) - \sqrt{\rho} \cdot X_M}{\sqrt{1 - \rho}}|X_M\right) \end{aligned}$$

This formula gives the relationship between the unconditional default probability $q(t)$ and the conditional default probability $q(t|X_M)$.

5.2 Homogeneous Portfolio

Under this assumption, all credit names have the same notional amount, recovery rate, and the credit curve. In this case, we just need to calculate the total number of defaults. Conditional on the common factor, default of one credit is independent from the default of another name. Then, the number of default is a Binominal distribution with the following density function:

$$P(N = k|X_M) = \binom{n}{k} q(t|X_M)^k (1 - q(t|X_M))^{n-k}, k=0, 1, 2, \dots, n \quad (5)$$

¹Suppose we have n assets whose pairwise correlation is ρ . Could ρ take a negative value? How negative value could it be?

where $q(t|X_M)$ is the conditional default probability before time t on the common factor.

But to obtain the unconditional number of defaults, we have to integrate the common factor out.

$$\begin{aligned}
P(N = k) &= \int \Pr(N = k|X_M) f(x_M) dx_M \\
&= \int \Pr(N = k|X_M) \frac{1}{\sqrt{2\pi}} e^{-\frac{x_M^2}{2}} dx_M \\
&= \frac{1}{\sqrt{\pi}} \int \Pr(N = k|\sqrt{2}x) e^{-x^2} dx \\
&= \frac{1}{\sqrt{\pi}} \sum_{i=1}^m \Pr(N = k|\sqrt{2}x_i) w_i
\end{aligned}$$

where (x_i, w_i) are Gaussian quadrature points, and m is the number of points chosen to obtain certain accuracy.

5.3 Conditional Normal Approximation

Conditional on the common factor X_M , we can see that the total loss as a random variable is simply the sum of independent random numbers which follows binary distribution. If the number is large, based on the *central limit theorem*, the total loss can be approximated by a normal distribution. Then, we simply need to know the conditional mean and the conditional variance.

$$\begin{aligned}
\mu(X_M) &= \sum_{i=1}^n N_i(1 - R_i) q_i(t|X_M) \\
\sigma^2(X_M) &= \sum_{i=1}^n [N_i(1 - R_i)]^2 q_i(t|X_M) (1 - q_i(t|X_M))
\end{aligned}$$

Then, the unconditional loss distribution can be obtained similarly as in the homogeneous portfolio case by integrating out the common factor. The issue with this approach is that there is non zero probability for the approximate loss distribution to have negative loss, which is not true, as loss could only take positive values.

5.4 Recursive Method

Conditional on the common factor the total loss is simply a sum of n independent random variables, that is

$$L_n(x|X_M) = Y_1(x|X_M) + Y_2(x|X_M) + \dots + Y_n(x|X_M)$$

where $Y_i(x|X_M)$ is $B_i = N_i(1 - R_i)$ if it defaults with probability $q_i(t|X_M)$ and zero if it does not. We also have this recursive formula for the conditional loss

$$L_j(x|X_M) = L_{j-1}(x|X_M) + Y_j(x|X_M), \quad (6)$$

for $j = 1, 2, \dots, n$. Hence, we can use *convolution* to directly calculate the total conditional loss distribution in a recursive way as follows:

$$f_{L_j}(x|X_M) = \begin{cases} (1 - q_j(t|X_M)) f_{L_{j-1}}(x|X_M), & \text{if } x < B_j \\ (1 - q_j(t|X_M)) f_{L_{j-1}}(x|X_M) + q_j(t|X_M) f_{L_{j-1}}(x - B_j|X_M), & \text{if } x \geq B_j \end{cases}$$

This approach is applicable for all cases, but in practice we need to have some approximation. For example, if all recovery adjusted notional amounts B_j 's are all different, then we need to have 2^n discrete points to represent the possible loss outcomes. For $n = 100$, that will be 1.2676505×10^{30} !

We very often try to divide the total maximum possible loss, which is $\sum_{i=1}^n B_i$ into some m equal bins with a bin size $\Delta h = \sum_{i=1}^n B_i / m$ and take each B_j just as a multiple of Δh , then we can simply calculate the probability masses on these fixed m points.

6 Tranche Loss Calculation

Suppose we have a total loss distribution for a n-name credit portfolio as $F(x)$, and a density function $f(x)$. A tranche expressed as $[K_L, K_U]$ is like an insurance contract on the total loss with deductible K_L and ceiling $K_U - K_L$.

The payoff function for the tranche $[K_L, K_U]$ is as follows:

$$L^T(t) = \begin{cases} 0, & L(t) < K_L \\ L(t) - K_L, & K_L \leq L(t) < K_U \\ K_U - K_L, & L(t) \geq K_U \end{cases}$$

or equivalently, $L^T(t) = \max[L(t) - K_L, 0] - \max[L(t) - K_U, 0]$.

Thus, we can view the tranche as a call spread on the total portfolio loss with strikes K_L and K_U .

Next, we derive the expected values:

$$E[\max(L(t) - K_L, 0)] = \int_{K_L}^{\infty} (1 - F(x)) dx$$

Similarly,

$$E[\max(L(t) - K_U, 0)] = \int_{K_U}^{\infty} (1 - F(x)) dx$$

Therefore, the expected loss transfer is:

$$E[L^T(t)] = \int_{K_L}^{K_U} (1 - F(x)) dx$$

Here, $1 - F(x)$ represents the *excess loss* distribution function. This shows that the expected tranche loss is simply the area below the excess loss distribution function bounded by the subordination levels².

The conditional tranche loss under the Gaussian approximation can be expressed as follows:

$$\begin{aligned} E(L^T(t)|X_M) &= (\mu_{X_M} - K_L)\Phi\left(\frac{\mu_{X_M} - K_L}{\sigma_{X_M}}\right) + \sigma_{X_M}\phi\left(\frac{\mu_{X_M} - K_L}{\sigma_{X_M}}\right) \\ &\quad - (\mu_{X_M} - K_U)\Phi\left(\frac{\mu_{X_M} - K_U}{\sigma_{X_M}}\right) - \sigma_{X_M}\phi\left(\frac{\mu_{X_M} - K_U}{\sigma_{X_M}}\right) \end{aligned}$$

The unconditional tranche loss can again be calculated using the Gaussian quadrature integration.

7 Conclusion

This document provides an overview of credit portfolio loss calculations with mathematical expressions in details.

References

- [1] Li, D. (2000). *On Default Correlation: A Copula Function Approach*. The Journal of Fixed Income, 9(4), 43-54.

²For any continuous positive random variable whose distribution function is $F(x)$, its expected value could be proven to be $\int_0^{\infty} (1 - F(x)) dx$